12.3 Causes of cancer

Some of the causes of cancer are:

1. Exposure to toxic compounds such as benzene, asbestos, nickel, cadmium, vinyl chloride, benzidine and N-nitrosamines.
2. Exposure to radiations from x-rays, radioactive materials and atomic exhausts.
3. Exposure to cigarette smoke, pesticide residues, water and air pollutants, synthetic food additives and contaminants.
4. Non-ionizing radiation from sunlight may damage the DNA of cells leading to cancer.
5. Tumor-producing viruses are called oncogenic viruses. Some of the viruses associated with human cancers are Human papillomavirus (HPV), EBV or Epstein-Barr virus, hepatitis viruses and Kaposi's sarcoma-associated herpes virus (KSHV).
6. Salted, smoked and processed meats contain nitrites and their related compounds which get converted to carcinogenic N-nitroso compounds in the colon.
7. Fat that drips from meat while being barbecued / grilled creates smoke that contains carcinogenic chemicals called **polycyclic aromatic hydrocarbons** which surrounds the meat. Another group of carcinogenic chemicals known as **hetero cyclic amines** are found in meat that has been fried or grilled for long periods of time.
8. Aflatoxins are toxins that are produced by certain fungi that grow on groundnuts. Ingestion of foods contaminated with aflatoxin is associated with cancer risk.
9. High-fat diets have been associated with cancer of the prostate gland, colon, breast, and uterus. By promoting obesity, a high sugar intake may indirectly increase cancer risk.
10. Individuals with reduced physical activity may be at an increased risk of developing cancer.

12.4 Symptoms of cancer

The American Cancer Society (2014) describes seven warning signs that a cancer may be present. They are:

- Changes in bowel or bladder habits
- A sore throat that does not heal
- Unusual bleeding or discharge
- Thickening or lump in the breast, testicles, or elsewhere
- Chronic indigestion and difficulty swallowing
- Noticeable change in the size, colour, shape, or thickness of a wart or mole
- Persistent cough or hoarseness.

Other signs may include the following:

- Loss of appetite
- Unexplained weight loss
- Persistent fatigue, nausea, or vomiting
- Unexplained low-grade fevers that may be either persistent or temporary



Oncology is a branch of medicine that deals with the prevention, diagnosis, and treatment of cancer. A doctor who specializes in treating cancer is called an **Oncologist**.

12.5 Stages of cancer

Staging describes the extent or spread of cancer at the time of diagnosis. Cancers fall between the stages of 0 to IV.

Stage 0: In this stage, cancers are still located in the place they started and have not spread to the surrounding tissues.

Stage I: It is also called e arly-stage cancer. This stage is usually a small cancer or tumor

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